INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE • Second Year

INTRODUCTION TO PROGRAMMING USING JAVA LAB

BCS351 • Complete Lab File

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Assignments 1–15

Submission note

This soft-copy file contains the Java source code and corresponding program output for every question in the supplied BCS351 lab-assignment sheet. The programs use fixed sample values from the assignment; no task in the supplied sheet requires console input, so separate input screenshots are not applicable.

The code panels are clean IDE-style renderings of the submitted Main.java source files, rather than photographs captured from a live IDE. Output panels show the generated output for each program.

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 1
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("Welcome to Java");
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
JAVA LAB - PROGRAM OUTPUT
-------------------------

HELLO WORLD
WELCOME TO JAVA
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 1
public class Main {
    public static void main(String[] args) {
        int roll = 101;
        double percentage = 84.5;
        char section = 'B';
        String name = "Asha";
        System.out.println("Roll: " + roll);
        System.out.println("Percentage: " + percentage);
        System.out.println("Section: " + section);
        System.out.println("Name: " + name);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
JAVA LAB - PROGRAM OUTPUT
-------------------------

ROLL: 101
PERCENTAGE: 84.5
SECTION: B
NAME: ASHA
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 1
public class Main {
    public static void main(String[] args) {
        int a = 17, b = 5;
        System.out.println("Sum: " + (a + b));
        System.out.println("Difference: " + (a - b));
        System.out.println("Product: " + (a * b));
        System.out.println("Quotient: " + (a / b));
        System.out.println("Remainder: " + (a % b));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
JAVA LAB - PROGRAM OUTPUT
-------------------------

SUM: 22
DIFFERENCE: 12
PRODUCT: 85
QUOTIENT: 3
REMAINDER: 2
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 4: Question4

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 1
public class Main {
    public static void main(String[] args) {
        final double PI = 3.14;
        double radius = 7;
        double area = PI * radius * radius;
        double perimeter = 2 * PI * radius;
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
JAVA LAB - PROGRAM OUTPUT
-------------------------

AREA: 153.86
PERIMETER: 43.96
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 2 | Question: 1
2  public class Main{public static void main(String[]x){int n=27;if(n%2==0)System.out.println(n+" is
3      even");else System.out.println(n+" is odd");}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
27 is odd
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 2 | Question: 2
2  public class Main{public static void main(String[]x){int
3      marks=83;if(marks>=90)System.out.println("Grade: A");else if(marks>=75)System.out.println("Grade:
4      B");else System.out.println("Grade: C");}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Grade: B
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 2 | Question: 3
2  public class Main{public static void main(String[]x){int
3      year=2024;if((year%4==0&&year%100!=0)||year%400==0)System.out.println("Leap year");else
4      System.out.println("Not a leap year");}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Leap year
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 4: Question4

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 2 | Question: 4
2  public class Main{public static void main(String[]x){int f=1;for(int
3      i=1;i<=5;i++)f*=i;System.out.println("Factorial of 5 is "+f);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Factorial of 5 is 120
```

Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 3 | Question: 1
2  public class Main{public static void main(String[]x){for(int i=1;i<=4;i++){for(int
3      j=1;j<=4;j++)System.out.print("*");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
****
****
****
****
```

Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 2: Question2

Code screenshot (IDE-style source view)

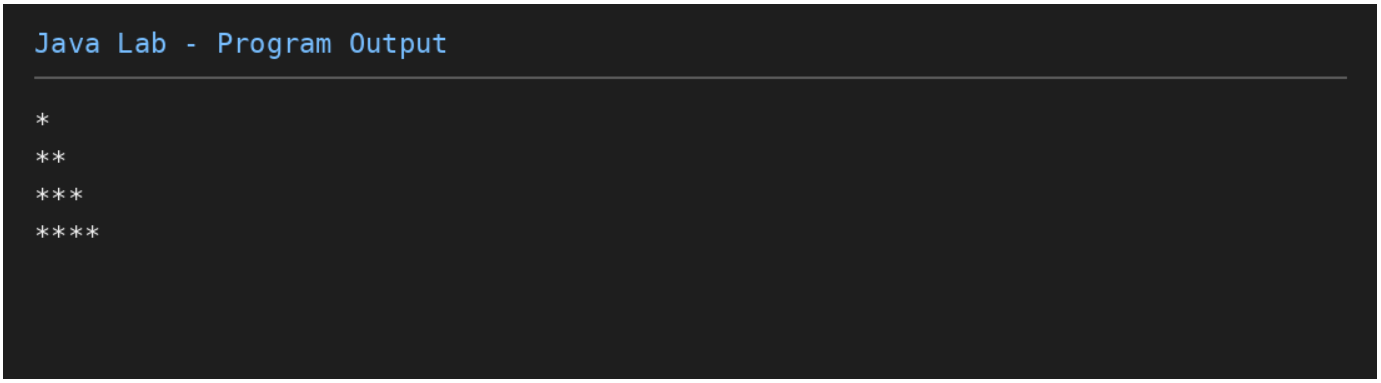
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 3 | Question: 2
2  public class Main{public static void main(String[]x){for(int i=1;i<=4;i++){for(int
3      j=1;j<=i;j++)System.out.print("*");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 3: Question3

Code screenshot (IDE-style source view)

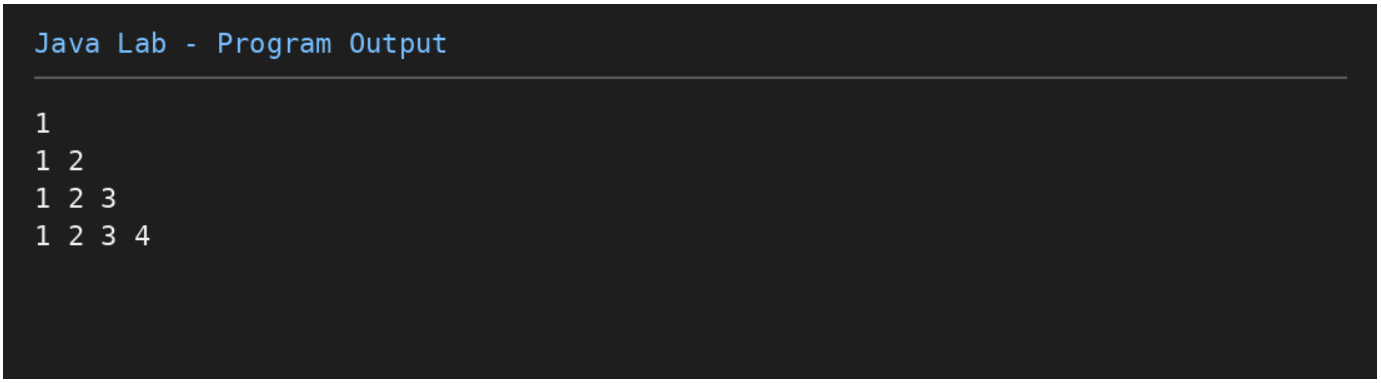
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 3 | Question: 3
2  public class Main{public static void main(String[]x){for(int i=1;i<=4;i++){for(int
3      j=1;j<=i;j++)System.out.print(j+" ");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 4: Question4

Code screenshot (IDE-style source view)

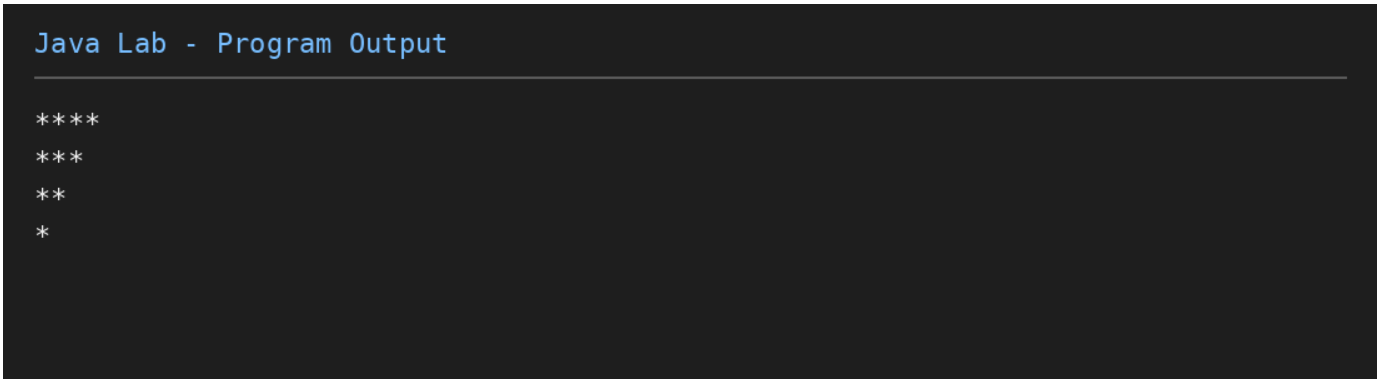
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 3 | Question: 4
2  public class Main{public static void main(String[]x){for(int i=4;i>=1;i--){for(int
3      j=1;j<=i;j++)System.out.print("*");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 4 | Question: 1
2  public class Main{static int fact(int n){int f=1;for(int i=1;i<=n;i++)f*=i;return f;}static int nCr(int
3      n,int r){return fact(n)/(fact(r)*fact(n-r));}public static void
4      main(String[]x){System.out.println("5C2 = "+nCr(5,2));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

5C2 = 10
```

Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 4 | Question: 2
2  public class Main{static int fact(int n){int f=1;for(int i=1;i<=n;i++)f*=i;return f;}static int nPr(int
3      n,int r){return fact(n)/fact(n-r);}public static void main(String[]x){System.out.println("5P2 =
4      "+nPr(5,2));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
5P2 = 20
```

Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 3: Question3

Code screenshot (IDE-style source view)

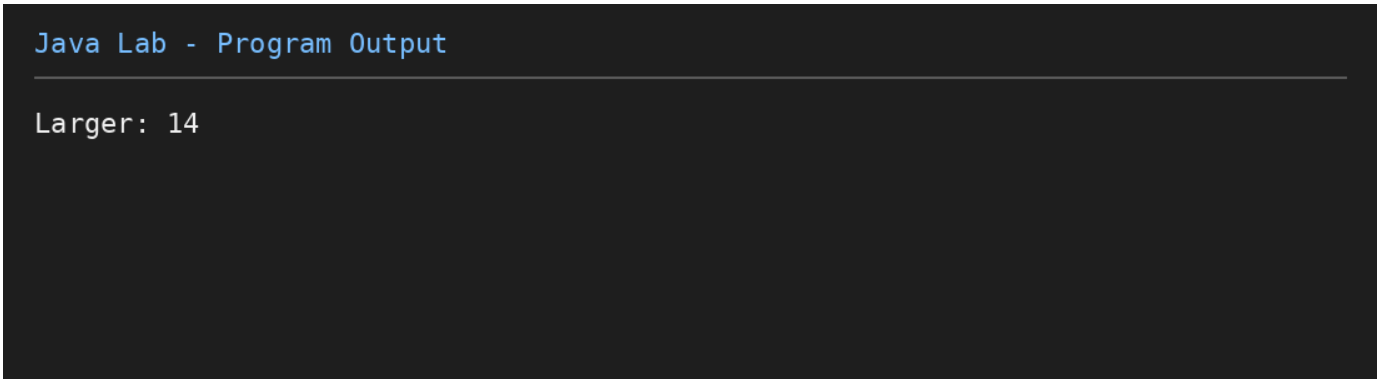
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 4 | Question: 3
2  public class Main{static int maxOf(int a,int b){return a>b?a:b;}public static void
3      main(String[]x){System.out.println("Larger: "+maxOf(14,9));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 5 | Question: 1
2  public class Main{public static void main(String[]x){int[] arr={3,4,5};for(int
3      i=0;i<arr.length;i++)System.out.println("Index "+i+": "+arr[i]);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Index 0: 3
Index 1: 4
Index 2: 5
```

Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 5 | Question: 2
2  public class Main{public static void main(String[]x){int[] a={12,45,7,33,21};int sum=0,max=a[0];for(int
3      v:a){sum+=v;if(v>max)max=v;}System.out.println("Sum: "+sum);System.out.println("Largest: "+max);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Sum: 118
Largest: 45
```

Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 5 | Question: 3
2  public class Main{public static void main(String[]x){int[] a={5,12,2,14,7};int c=0;for(int
3      v:a)if(v>10)c++;System.out.println("Count greater than 10: "+c);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

Count greater than 10: 2
```

Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 1: Question1

Code screenshot (IDE-style source view)

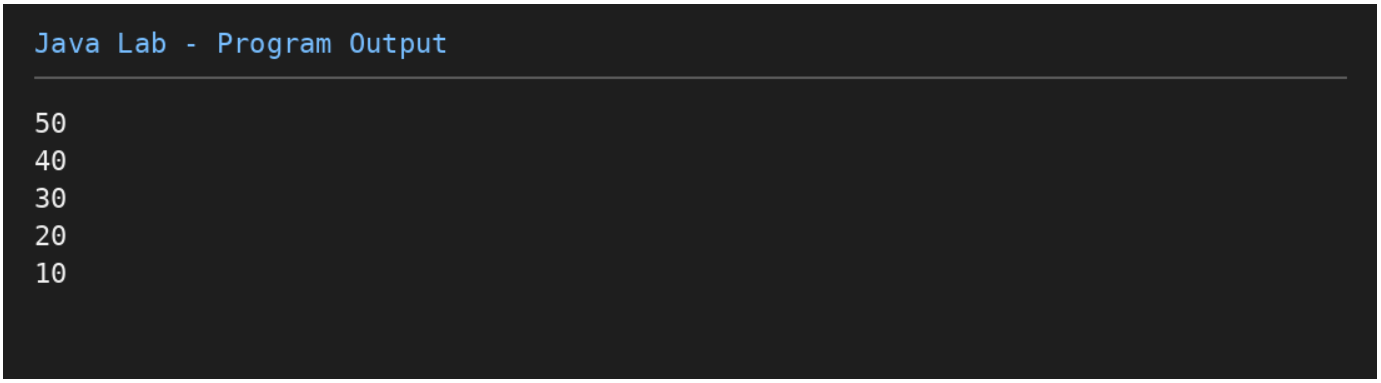
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 6 | Question: 1
2  public class Main{public static void main(String[]x){int[] a={10,20,30,40,50};for(int
3      i=a.length-1;i>=0;i--)System.out.println(a[i]);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 2: Question2

Code screenshot (IDE-style source view)

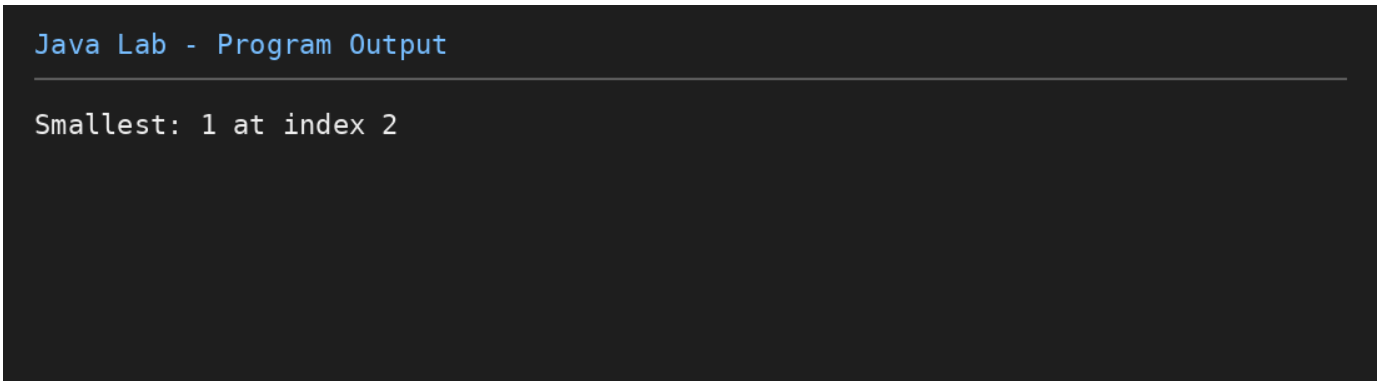
Main.java | IDE-style code view

```
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 6 | Question: 2
public class Main{public static void main(String[]x){int[] a={2,5,1,8,3,4};int min=a[0],idx=0;for(int
    i=1;i<a.length;i++)if(a[i]<min){min=a[i];idx=i;}System.out.println("Smallest: "+min+" at index
    "+idx);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 3: Question3

Code screenshot (IDE-style source view)

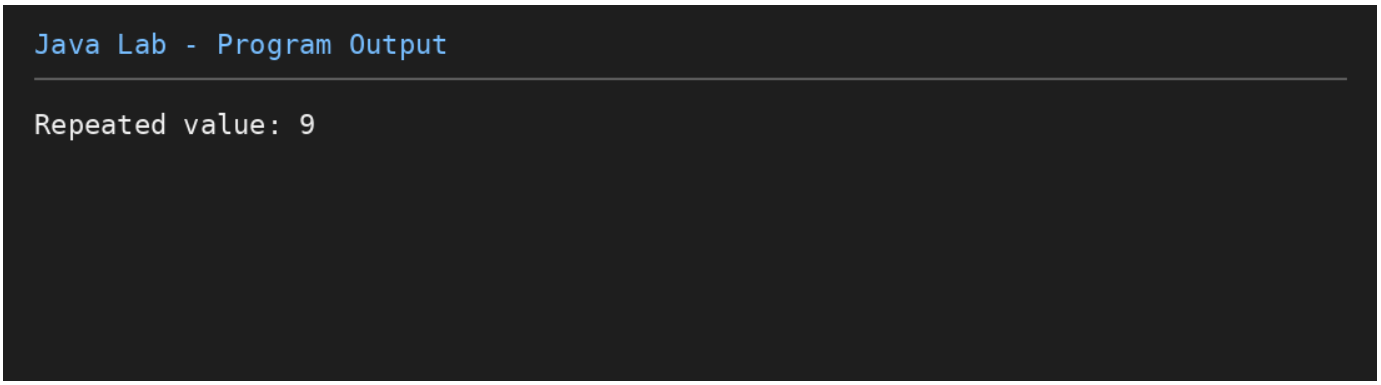
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 6 | Question: 3
2  public class Main{public static void main(String[]x){int[] a={4,9,2,9,7};boolean found=false;for(int
3      i=0;i<a.length&&!found;i++)for(int j=i+1;j<a.length;j++)if(a[i]==a[j]){System.out.println("Repeated
4      value: "+a[i]);found=true;break;}if(!found)System.out.println("No repeated value");}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 7 | Question: 1
public class Main{public static void main(String[]x){int steps=0;// Time complexity: O(n)
for(int i=1;i<=10;i++)steps++;System.out.println("Steps: "+steps);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

Steps: 10
```

Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 2: Question2

Code screenshot (IDE-style source view)

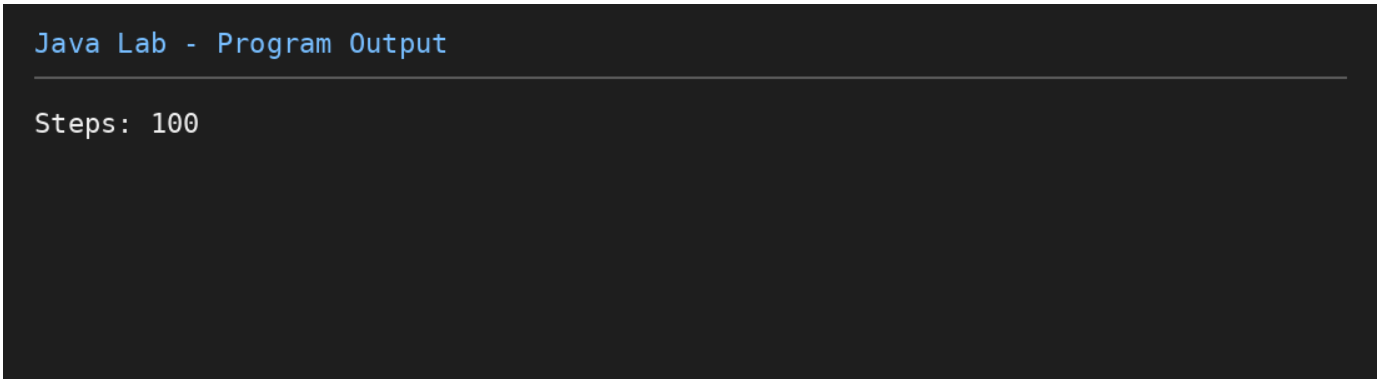
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 7 | Question: 2
2  public class Main{public static void main(String[]x){int steps=0;// Time complexity: O(n^2)
3  for(int i=1;i<=10;i++)for(int j=1;j<=10;j++)steps++;System.out.println("Steps: "+steps);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 3: Question3

Code screenshot (IDE-style source view)

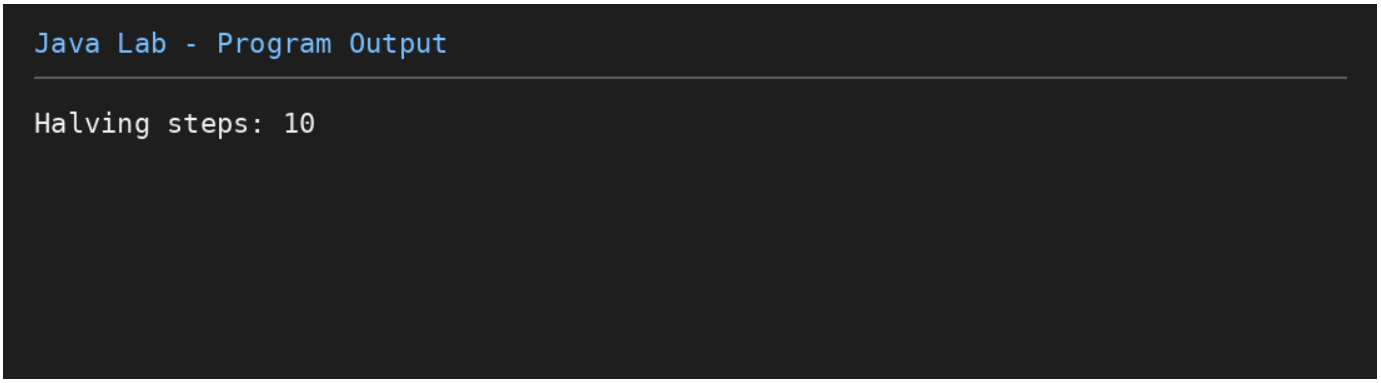
Main.java | IDE-style code view

```java
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 7 | Question: 3
public class Main{public static void main(String[]x){int n=1024,steps=0;// Time complexity: O(log n)
while(n>1){n/=2;steps++;}System.out.println("Halving steps: "+steps);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 8 | Question: 1
2  public class Main{public static void main(String[]x){int[][] m=new int[3][4];int v=1;for(int
3      i=0;i<3;i++)for(int j=0;j<4;j++)m[i][j]=v++;for(int[] row:m){for(int a:row)System.out.print(a+"
4      ");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
1 2 3 4
5 6 7 8
9 10 11 12
```

Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 2: Question2

Code screenshot (IDE-style source view)

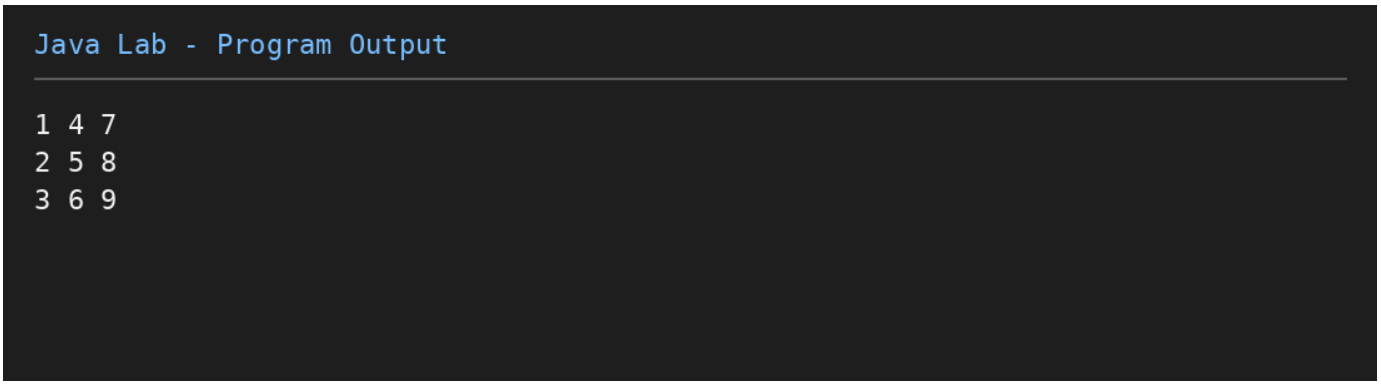
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 8 | Question: 2
2  public class Main{public static void main(String[]x){int[][] m={{1,2,3},{4,5,6},{7,8,9}},t=new
3      int[3][3];for(int i=0;i<3;i++)for(int j=0;j<3;j++)t[j][i]=m[i][j];for(int[] row:t){for(int
4      a:row)System.out.print(a+" ");System.out.println();}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 3: Question3

Code screenshot (IDE-style source view)

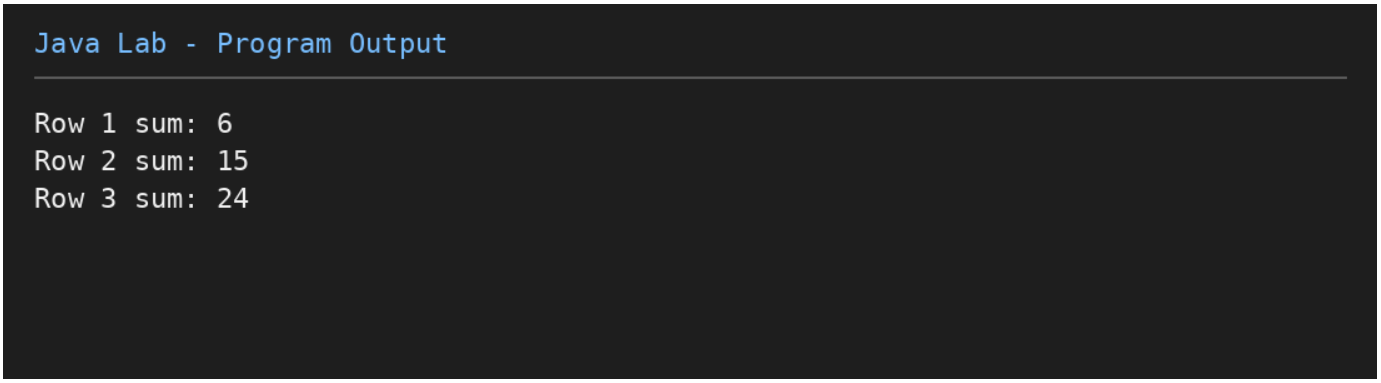
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 8 | Question: 3
2  public class Main{public static void main(String[]x){int[][] m={{1,2,3},{4,5,6},{7,8,9}};for(int
3      i=0;i<3;i++){int s=0;for(int j=0;j<3;j++)s+=m[i][j];System.out.println("Row "+(i+1)+" sum: "+s);}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 9 | Question: 1
2  public class Main{public static void main(String[]x){int[][]
3      m={{1,2,3},{4,5,6},{7,8,9}};System.out.print("First row: ");for(int v:m[0])System.out.print(v+"
4      ");System.out.println();System.out.print("Last row: ");for(int v:m[2])System.out.print(v+"
5      ");System.out.println();}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
First row: 1 2 3
Last row: 7 8 9
```

Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 9 | Question: 2
2  public class Main{public static void main(String[]x){int[][] m={{1,2,3},{4,5,6},{7,8,9}};for(int
3      j=0;j<3;j++){int s=0;for(int i=0;i<3;i++)s+=m[i][j];System.out.println("Column "+(j+1)+" sum:
4      "+s);}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Column 1 sum: 12
Column 2 sum: 15
Column 3 sum: 18
```

Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 9 | Question: 3
2  public class Main{public static void main(String[]x){int[][] m={{1,2,3},{4,5,6},{7,8,9}};int
3      c=0;for(int[] row:m)for(int v:row)if(v>5)c++;System.out.println("Values greater than 5: "+c);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

Values greater than 5: 4
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 10 | Question: 1
2  public class Main{public static void main(String[]x){int day=3;switch(day){case
3      1:System.out.println("Monday");break;case 2:System.out.println("Tuesday");break;case
4      3:System.out.println("Wednesday");break;default:System.out.println("Invalid day");}}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Wednesday
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 10 | Question: 2
2  import java.util.ArrayList; public class Main{public static void main(String[]x){ArrayList<Integer>
3      list=new ArrayList<>();list.add(10);list.add(20);list.add(30);list.add(40);System.out.println(list);
4      System.out.println("Size: "+list.size());System.out.println("Item at index 2: "+list.get(2));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
[10, 20, 30, 40]
Size: 4
Item at index 2: 30
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 10 | Question: 3
2  import java.util.ArrayList; public class Main{public static void main(String[]x){ArrayList<Integer>
3      list=new ArrayList<>();list.add(10);list.add(20);list.add(30);list.add(40);list.set(1,25);list.remov
4      e(3);System.out.println(list);System.out.println("Size: "+list.size());}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
[10, 25, 30]
Size: 3
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 11 | Question: 1
2  import java.util.Arrays; public class Main{public static void main(String[]x){int[]
3      a={5,1,4,2,8};for(int i=0;i<a.length-1;i++)for(int j=0;j<a.length-1-i;j++)if(a[j]>a[j+1]){int
4      t=a[j];a[j]=a[j+1];a[j+1]=t;}System.out.println("Sorted: "+Arrays.toString(a));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

Sorted: [1, 2, 4, 5, 8]
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 11 | Question: 2
2  import java.util.Arrays; public class Main{public static void main(String[]x){int[]
3      a={64,25,12,22,11};for(int i=0;i<a.length-1;i++){int min=i;for(int
4      j=i+1;j<a.length;j++)if(a[j]<a[min])min=j;int
5      t=a[i];a[i]=a[min];a[min]=t;}System.out.println("Sorted: "+Arrays.toString(a));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Sorted: [11, 12, 22, 25, 64]
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 11 | Question: 3
2  public class Main{public static void main(String[]x){int[] a={5,1,4,2,8};int swaps=0;for(int
3      i=0;i<a.length-1;i++)for(int j=0;j<a.length-1-i;j++)if(a[j]>a[j+1]){int
4      t=a[j];a[j]=a[j+1];a[j+1]=t;swaps++;}System.out.println("Swaps: "+swaps);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Swaps: 4
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 12 | Question: 1
2  import java.util.Arrays; public class Main{public static void main(String[]x){int[] a={7,3,5,1};for(int
3      i=1;i<a.length;i++){int
4      key=a[i],j=i-1;while(j>=0&&a[j]>key){a[j+1]=a[j];j--;}a[j+1]=key;}System.out.println("Sorted:
5      "+Arrays.toString(a));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Sorted: [1, 3, 5, 7]
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 12 | Question: 2
import java.util.Arrays; public class Main{public static void main(String[]x){int[]
    a={14,3,9,21,6};for(int i=1;i<a.length;i++){int
    key=a[i],j=i-1;while(j>=0&&a[j]>key){a[j+1]=a[j];j--;}a[j+1]=key;}System.out.println("Sorted:
    "+Arrays.toString(a));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Sorted: [3, 6, 9, 14, 21]
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 3: Question3

Code screenshot (IDE-style source view)

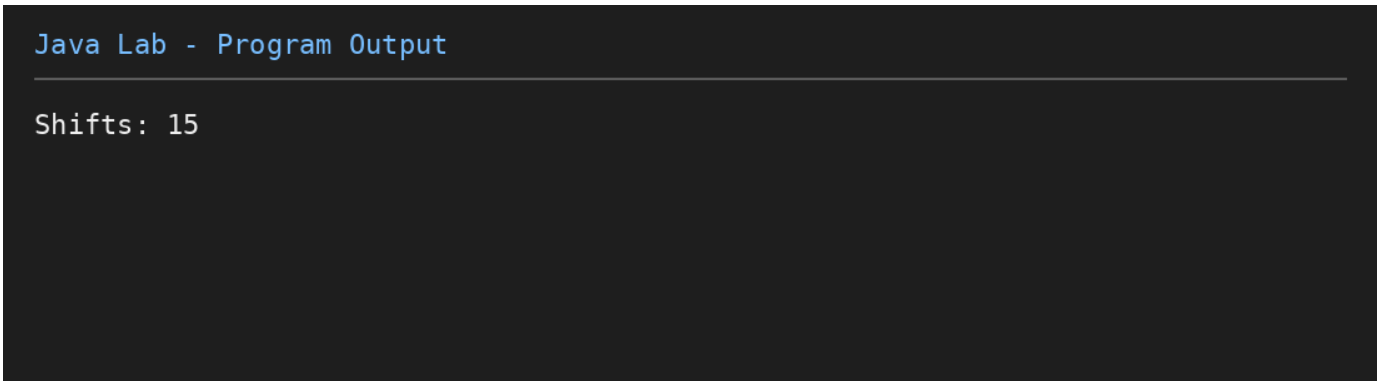
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 12 | Question: 3
2  public class Main{public static void main(String[]x){int[] a={6,5,4,3,2,1};int shifts=0;for(int
3      i=1;i<a.length;i++){int key=a[i],j=i-1;while(j>=0&&a[j]>key){a[j+1]=a[j];j--;shifts++;}a[j+1]=key;}S
4      ystem.out.println("Shifts: "+shifts);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
// Name: Ayushi | Roll No.: BCS2025414 | Assignment: 13 | Question: 1
public class Main{public static void main(String[]x){String s="Programming";System.out.println("Length:
    "+s.length());System.out.println("Position of g: "+s.indexOf('g'));System.out.println("Upper case:
    "+s.toUpperCase());}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Length: 11
Position of g: 3
Upper case: PROGRAMMING
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 2: Question2

Code screenshot (IDE-style source view)

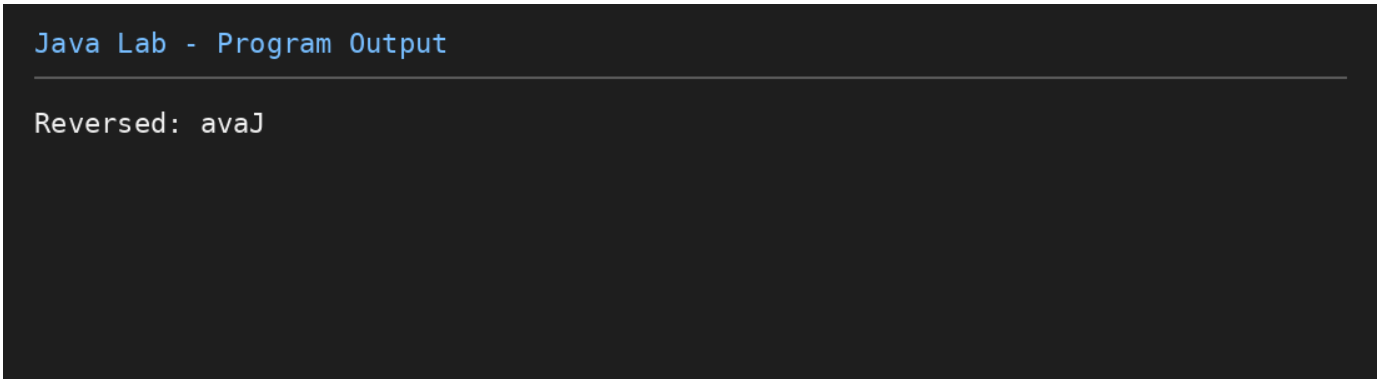
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 13 | Question: 2
2  public class Main{public static void main(String[]x){String s="Java",r="";for(int
3      i=s.length()-1;i>=0;i--)r+=s.charAt(i);System.out.println("Reversed: "+r);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 13 | Question: 3
2  public class Main{public static void main(String[]x){String s="racecar",r="";for(int
3      i=s.length()-1;i>=0;i--)r+=s.charAt(i);if(s.equals(r))System.out.println(s+" is a palindrome");else
4      System.out.println(s+" is not a palindrome");}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
racecar is a palindrome
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 4: Question4

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 13 | Question: 4
2  public class Main{public static void main(String[]x){String s="abc";for(int i=0;i<s.length();i++)for(int
3      j=i+1;j<=s.length();j++)System.out.println(s.substring(i,j));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
a
ab
abc
b
bc
c
```

Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 1: Question1

Code screenshot (IDE-style source view)

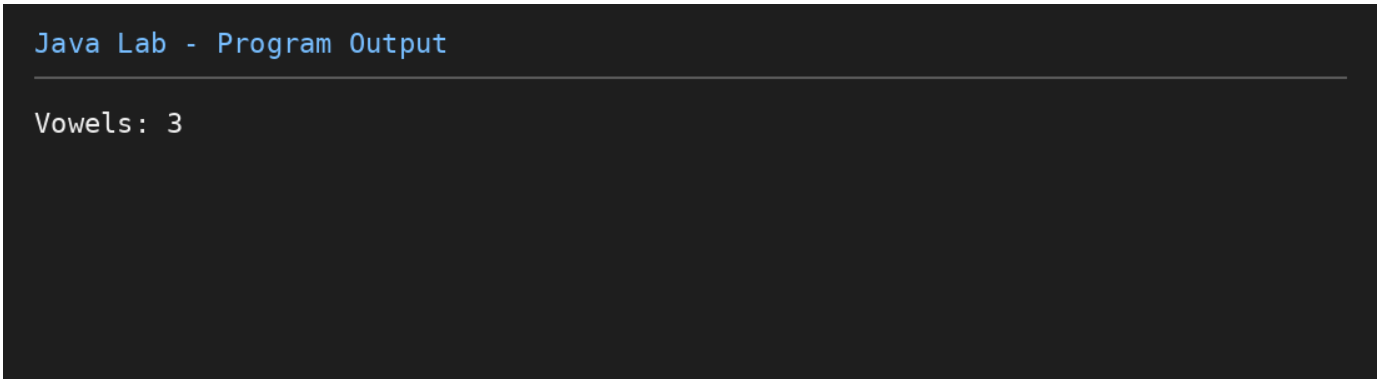
Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 14 | Question: 1
2  public class Main{public static void main(String[]x){String s="programming";int c=0;for(int
3      i=0;i<s.length();i++)if("aeiou".indexOf(s.charAt(i))>=0)c++;System.out.println("Vowels: "+c);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 14 | Question: 2
2  public class Main{public static void main(String[]x){String s="programming";int c=0;for(int
3      i=0;i<s.length();i++)if(s.charAt(i)=='g')c++;System.out.println("Count of g: "+c);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Count of g: 2
```

Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 14 | Question: 3
2  public class Main{public static void main(String[]x){String s="hello";for(int
3      i=0;i<s.length();i++)System.out.println(i+": "+s.charAt(i));}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
0: h
1: e
2: l
3: l
4: o
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 15 | Question: 1
2  public class Main{public static void main(String[]x){StringBuilder sb=new
3      StringBuilder("Hello");sb.append(" World");System.out.println(sb);System.out.println("Length:
4      "+sb.length());}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
Hello World
Length: 11
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 15 | Question: 2
2  public class Main{public static void main(String[]x){StringBuilder sb=new StringBuilder();for(int
3      i=1;i<=5;i++)sb.append(i).append(',');sb.deleteCharAt(sb.length()-1);System.out.println(sb);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

Java Lab - Program Output

```
1,2,3,4,5
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java | IDE-style code view

```
1  // Name: Ayushi | Roll No.: BCS2025414 | Assignment: 15 | Question: 3
2  public class Main{public static void main(String[]x){StringBuilder sb=new
3      StringBuilder("Java");sb.insert(0,"Hello ");System.out.println(sb);}}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
Java Lab - Program Output

Hello Java
```